

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 2

Year: 2019

Subject: 392152 Chemistry II

Section: 15-18

Date: 4 March 2020

Time: 10.00-12.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to evaluate your comprehension. The test is divided into 2 parts. There will be 9 pages (including this page) and they are worth 30 points.

- Answer the questions on this test paper.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators are **NOT** allowed in the examination.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

	Multiple Choice	Part 2(1)	Part 2(2)	Total
score				

Part 1. Multiple Choice (10 points)

Choose the letter of the choice in the answer table that best completes the statement or answers the question.

Question no.	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

1. The limiting reagent in a chemical reaction is one that

- (a) has the smallest molar mass (formula weight).
- (b) has the smallest coefficient.
- (c) is consumed completely.
- (d) is in excess.

2. Hydrogen and nitrogen react to form ammonia according to the reaction, $\text{H}_2 + \text{N}_2 \rightarrow \text{NH}_3$.

If 4.0 mol of H_2 with 2.0 mol of N_2 are reacted, how many moles of what reactant is left over?

- (a) 0.7 mol of N_2
- (b) 1.3 mol of N_2
- (c) 2 mol of H_2
- (d) 6 mol of H_2

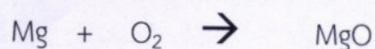
3. Zinc metal reacts with hydrochloric acid by the following reaction:



how many moles of ZnCl_2 are produced if 32.75 g of Zn is added to excess HCl solution?

- (a) 0.5 mol
- (b) 1.0 mol
- (c) 1.5 mol
- (d) 2.0 mol

4. Magnesium and oxygen gas react to form magnesium oxide as the following equation.



If 24.0 g of magnesium and 32.0 g of oxygen are mixed, which statement is incorrect?

- (a) Mg is a limiting reagent.
- (b) O₂ is an excess reactant.
- (c) 0.5 mol MgO will be formed.
- (d) The remaining of an excess reactant is 16 g.

5. Chlorine is produced by the reaction $2\text{HCl(g)} \rightarrow \text{H}_2\text{(g)} + \text{Cl}_2\text{(g)}$. How many grams of HCl must be used to produce 11.2.0 L of chlorine at STP?

- (a) 9.12 g
- (b) 36.50 g
- (c) 18.25 g
- (d) 73.00 g

6. Which statement about noble gases is correct?

- (a) They form compounds with very bright colors.
- (b) They exist as single atoms rather than as molecules.
- (c) They are highly reactive with both metals and nonmetals.
- (d) They are extremely rare in nature.

7. An atom with an atomic radius smaller than that of sulfur (S) is _____.

- (a) Chlorine (Cl)
- (b) Calcium (Ca)
- (c) Lithium (Li)
- (d) Potassium (K)

8. Which statement is true?

- (a) Barium has higher electronegativity than nitrogen.
- (b) Cerium has lower ionization energy than sodium.
- (c) Size of lithium and lithium ion is the same.
- (d) Chlorine ion is smaller than bromine ion.

9. Oxides of most nonmetals combine with water to form _____.

- (a) an acid
- (b) a base
- (c) water and a salt
- (d) water

10. Which statement is incorrect about alkaline metal reacting with water?

- (a) The solution will be a basic solution if it reacts with water.
- (b) One mole of H₂ will be formed.
- (c) It reacts with water with the ratio of 1:1.
- (d) The reaction produces heat.

Part 2 Write your answer in the space provided for each question.

Instruction

READ carefully

Students must provide calculation detail if CALCULATION is noted for any questions. Getting a full score cannot be provided if only answer without any detail is written.

Part 2.1 Limiting reagent (5 points) and periodic table trend (5 points)

1. Determine the limiting reagent if

(CALCULATION, 3 points)

a. 1.4 mol Al and 3.2 mol I₂ are used in the reaction : $\text{Al} + \text{I}_2 \rightarrow \text{AlI}_3$

b. 2.1 g propane (C₃H₈) reacts with 3.2 g oxygen in the reaction:



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2. The reaction of iron ore with carbon follows the equation: $\text{Fe}_2\text{O}_3 + \text{C} \rightarrow \text{Fe} + \text{CO}_2$
- How many grams of Fe can be produced from a mixture of 80 g of Fe_2O_3 and 60 g of C?
 - How much CO_2 can theoretically form? (CALCULATION, 2 points)

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3. The atomic numbers of elements M, N and P are 14, 17 and 11, respectively. Complete the table below and answer the following questions. (2.5 points)

Element	Z	Electron configuration		Group	Period
		main	Subshell (noble gas can be used)		
M	14				
N	17				
P	11				

- a. Arrange the elements in order of increasing atomic radius.

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- b. Which element has highest first ionization energy? Provide reasons.

c. Which element has highest third ionization energy? Provide reasons.

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d. Arrange the following elements in order of increasing electronegativity.

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4. The ionization energies (kJ/mol) for three third period elements are given in the table below. Identify each of the elements.

	IE ₁	IE ₂	IE ₃	IE ₄	IE ₅	IE ₆	Element?
A	1,012	1,903	2,912	4,957	6,274	21,269	
B	496	4,562	6,912	9,544	13,353	16,610	
C	578	1,817	2,745	11,578	14,831	23,295	

a. Which element is a alkaline?

(0.5 points)

.....

b. Which elements should have -3 charges as its' ion?

(0.5 points)

.....

c. Which elements should a non-metal?

(0.5 points)

.....

d. Write the chemical equation of the fifth ionization energy of element C and how much energy (in kJ) must be used to remove 5 electrons of element C. (1 point)

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Part 2.2 Yield percentage (5 points) and Chloride and oxide compounds (5 points)

1. To prepare sodium sulphate (Na_2SO_4), the following reaction could be used.



What is the % yield if you use 5 g of NaOH and make 6 g of Na_2SO_4 ?

(CALCULATION, 3 points)

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2. Chloroform (CHCl_3) reacts with chlorine (Cl_2) to form carbon tetrachloride (CCl_4) and hydrogen chloride (HCl). In an experiment 25 grams of chloroform and 25 grams of chlorine were mixed. What is the maximum yield of CCl_4 in moles and in grams?

(CALCULATION, 2 points)

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3. Use the scientific words and fill in the blank properly of each sentence below. (3 points)

acidic oxide

basic oxide

amphoteric oxide

neutral oxide

water-soluble

water-insoluble

ionic

covalent

3.1 Sulfur dioxide isoxide compound.

3.2 Lithium chloride is.....

3.3bond can be seen in chlorine molecule.

3.4 Chemical bonding between carbon atoms and four chlorine atoms in

carbontetrachloride is notbond.

3.5is a compound that is not acidic and basic.

3.6 Al_2O_3 can be categorized aswhich is one of the oxide compounds.

4. Give a list of two neutral acid oxides and two acid oxides. (2 points)

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Nathjanan Jongkon and Sunanta Chuayprakong

IUPAC Periodic Table of the Elements

1		2		Key:																18											
1 H hydrogen 1.008		2 He helium 4.0026		atomic number Symbol name conventional atomic weight standard atomic weight																											
3	Li lithium 6.941	4	Be beryllium 9.0122																	10	Ne neon 20.180										
11	Na sodium 22.990	12	Mg magnesium 24.305																	17	F fluorine 18.998										
19	K potassium 39.098	20	Ca calcium 40.078(4)																	36	Kr krypton 83.798(2)										
37	Rb rubidium 85.468	38	Sr strontium 87.62																	54	Xe xenon 131.29										
55	Cs caesium 132.91	56	Ba barium 137.33																	86	Rn radon 222										
87	Fr francium	88	Ra radium																	118	Og oganesson										
21 Sc scandium 44.956		22 Ti titanium 47.867		23 V vanadium 50.942		24 Cr chromium 51.996		25 Mn manganese 54.938		26 Fe iron 55.845(2)		27 Co cobalt 58.933		28 Ni nickel 58.693		29 Cu copper 63.546(3)		30 Zn zinc 65.38(2)		31 Ga gallium 69.723		32 Ge germanium 72.630(8)		33 As arsenic 74.922		34 Se selenium 78.971(8)		35 Br bromine 79.904		36 Kr krypton 83.798(2)	
39 Y yttrium 88.906		40 Zr zirconium 91.224(2)		41 Nb niobium 92.906		42 Mo molybdenum 95.95		43 Tc technetium		44 Ru ruthenium 101.07(2)		45 Rh rhodium 102.91		46 Pd palladium 106.42		47 Ag silver 107.87		48 Cd cadmium 112.41		49 In indium 114.82		50 Sn tin 118.71		51 Sb antimony 121.76		52 Te tellurium 127.60(3)		53 I iodine 126.90		54 Xe xenon 131.29	
57-71 lanthanoids		72 Hf hafnium 178.49(2)		73 Ta tantalum 180.95		74 W tungsten 183.84		75 Re rhenium 186.21		76 Os osmium 190.23(3)		77 Ir iridium 192.22		78 Pt platinum 195.08		79 Au gold 196.97		80 Hg mercury 200.59		81 Tl thallium 204.38		82 Pb lead 207.2		83 Bi bismuth 208.98		84 Po polonium		85 At astatine		86 Rn radon	
89-103 actinoids		104 Rf rutherfordium		105 Db dubnium		106 Sg seaborgium		107 Bh bohrium		108 Hs hassium		109 Mt meitnerium		110 Ds darmstadtium		111 Rg roentgenium		112 Cn copernicium		113 Nh nihonium		114 Fl flerovium		115 Mc moscovium		116 Lv livermorium		117 Ts tennessine		118 Og oganesson	



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57	La	lanthanum	138.91
58	Ce	cerium	140.12
59	Pr	praseodymium	140.91
60	Nd	neodymium	144.24
61	Pm	promethium	
62	Sm	samarium	150.36(2)
63	Eu	euroium	151.96
64	Gd	gadolinium	157.25(3)
65	Tb	terbium	158.93
66	Dy	dysprosium	162.50
67	Ho	holmium	164.93
68	Er	erbium	167.26
69	Tm	thulium	168.93
70	Yb	ytterbium	173.05
71	Lu	lutetium	174.97
89	Ac	actinium	227.04
90	Th	thorium	232.04
91	Pa	protactinium	231.04
92	U	uranium	238.03
93	Np	neptunium	
94	Pu	plutonium	
95	Am	americium	
96	Cm	curium	
97	Bk	berkelium	
98	Cf	californium	
99	Es	einsteinium	
100	Fm	fermium	
101	Md	mendelevium	
102	No	nobelium	
103	Lr	lawrencium	

For notes and updates to this table, see www.iupac.org. This version is dated 1 December 2018.
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United Nations
Educational, Scientific and
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of the Periodic Table
of Chemical Elements

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